

① $p \wedge q$ es contradictorio
negación

Entonces $\rightarrow$ $p \wedge q$ es tautológico

$p \wedge q = \bar{p} \vee \bar{q}$ es cierto si $\bar{p}$ es verdadero
si y solo si $\bar{q}$ es verdadero
de Morgan

$\bar{p}$ es verdadero si p falso ; $\bar{q}$ es verdadero si q es falso
Por tanto la opción correcta es la (a).

② $p(x) : \forall y \in \mathbb{R}, xy < y^2$ ($y=0$) Fals

x	(a)	Sc counter	$p(0)$	$0 < y^2 \quad \forall y \in \mathbb{R} \quad (y=0 \text{ fals})$
x	(b)	"	$p(1)$	$y < y^2$
x	(c)	"	$p(-1)$	$\downarrow$ $-y < y^2$
✓	(d)	$\forall x \in \mathbb{R}$	$\overline{p(x)}$	$y = \frac{1}{2} > y^2 = \frac{1}{4}$ $y = -\frac{1}{2} \quad y^2 = \frac{1}{4}$ $-y = \frac{1}{2} > \frac{1}{4}$

$$xy < y^2 \quad (\Rightarrow) \quad 0 < y(y-x)$$

Note for $y=0$ $\forall x \in \mathbb{R} \quad \therefore \quad y=0$

Also

$$0 < \underbrace{0(0-x)}_{=0}$$

③ $|A| = 47$ $<$ $|B| = 53$, $f: \underline{A} \rightarrow B$ aplicación

f injectiva $x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$

(a)

$\circ f(x_1) = f(x_2) (\Rightarrow) x_1 = x_2$

Nunca

Correcta

$A = \{ \underline{a_1, a_2, \dots, a_{47}} \}$ $B = \{ \underline{b_1, b_2, \dots, b_{53}} \}$

$f(a_n) = b_n$



(b) f sobreyectiva $f(A) = B$

Recordar que f es signa
 e cada $x \in A$ uno y solo
 un $y \in B$ $|f(A)| = 47 < 53$

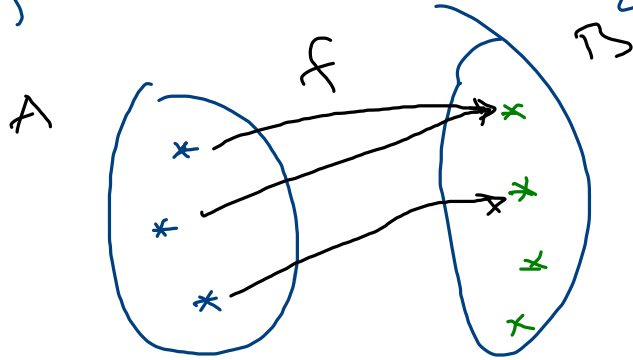
$$f(x) = b, \quad \forall x \in A$$

$$f(A) = \{b\}$$

$$|f(A)| \leq |A| < |B|$$

↑

f podría ser sobreyectiva sólo si $|B| \leq |A|$



Note $|f(A)| = |A|$
 i.e. f is injective.

f puede ser inyección

PERO no sobreyección $|A| < |B|$

Por $\therefore$ no puede ser bijcción inyección
&
~~sobreyección~~

(c) es cierto también

(a) f ~~tiene~~ inversa $\Rightarrow f$ es ~~bijcción~~
ser cierto (b), (c) y (d)

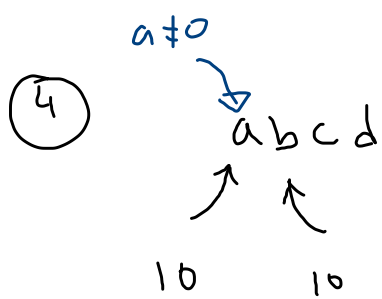


Diagram illustrating the calculation of the number of 4-digit numbers. It shows the expression $10^4 = 10.000$ in a box, followed by an arrow pointing to a box containing 10000 , and finally an arrow pointing to the result 9000 .

Adición: ¿Cuántos números de 4 cifras se pueden formar, en la digital del '0' al '9'?

$$9 \cdot 10 \cdot 10 \cdot 10 = 9000$$

$$\textcircled{5} \quad f(x) = \begin{cases} \sqrt{x^2 + 1} & x < 0 \\ \rightarrow 0 & x = 0 \\ \ln(x+1) + e^x & x > 0 \end{cases} \quad \left| \begin{array}{l} \textcircled{L} \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \underbrace{\ln(x+1) + e^x}_{= \ln(1) + e^0 = 0 + 1 = \underline{1}} \end{array} \right.$$

$$\textcircled{1} \quad \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \sqrt{x^2 + 1} = \sqrt{0+1} = \sqrt{1} = \underline{1} \neq f(0)$$

$$\left(\exists \lim_{x \rightarrow a} f(x) = L \iff \underbrace{\exists \lim_{x \rightarrow a^\pm} f(x) = \textcircled{L}} \right) \quad \left| \begin{array}{l} f \notin C(0) \therefore \\ f \notin D(0) \end{array} \right.$$

$$\textcircled{6} \quad p(A) = 0.4 \quad \& \quad p(B) = 0.5$$

$A \& B$ independent $\textcircled{*}$

$$p(A \cup B) = p(A) + p(B) - \underbrace{p(A \cap B)}_{\substack{11 \textcircled{*} \\ p(A)p(B)}}$$

$$\begin{aligned} p(A \cup B) &= 0.4 + 0.5 - 0.4 \cdot 0.5 \\ &= 0.9 - 0.2 = 0.7 \end{aligned}$$

$$\boxed{0.7}$$

$$\frac{4}{10} \cdot \frac{5}{10} = \frac{20}{100} = 0.20$$

⑦ $C(n) = \underbrace{(n^5 - 2n + 1)^2}_{(n^5 + (-2n + 1))^2}$ (# de operaciones realizadas $\perp$)

$$= n^{10} + 2(-2n + 1)n^5 + (-2n + 1)^2$$

$$= n^{10} - 4n^6 + 2n^5 + 4n^2 - 4n + 1$$

$$C(n) \sim \underbrace{O(n^{10})}_{(c)}$$

$$C(n) \sim O(n^k)$$

$$\lim_{n \rightarrow \infty} \frac{C(n)}{n^k} = l \neq 0 \quad k =$$

$$\lim_{n \rightarrow \infty} \frac{n^{10} - 4n^6 + 2n^5 + 4n^2 - 4n + 1}{n^{10}} = 1$$

$$a_n \neq 0 \quad \lim_{x \rightarrow \pm\infty} \frac{a_n x^n + \dots + a_1 x + a_0}{b_m x^m + \dots + b_1 x + b_0} = \lim_{x \rightarrow \pm\infty} \frac{a_n x^n}{b_m x^m} = \frac{a_n}{b_m} \quad (n=m)$$

$$b_m \neq 0$$

can we also say $b_m = 1$ & $n = m = 10$ ($a_n = 1$)

$$\textcircled{8} \quad A, B \in \mathbb{R}^{3 \times 3} \quad \det(A) = 2$$

$$\sigma(B) = \{-2, 1, 3\} \quad (\text{aufzuheben 20 B})$$

$$\text{Zu a) } \det(2A^{-1}B) = 2^3 \cdot \frac{1}{2} \cdot (-2 \cdot 1 \cdot 3) = \underline{\underline{-24}}$$

$$\det(\lambda A) = \lambda^n \det(A) \quad A \in \mathbb{R}^{n \times n}$$

$$\det(AB) = \det(A) \det(B) \quad \left| \quad \det(B) = \prod_{x \in \sigma(B)} x \right.$$

$$\det(A^{-1}) = 1 / \det(A)$$

① $A \in \mathbb{R}^{n \times n}$ symmetrisch

def. posit.

$\exists v \neq 0$ $Av = -2v$ gilt $\forall v \neq 0$

$-2 \in \sigma(A)$ (Eigenwert)
 $\Leftrightarrow$ Eigenwert von A

$\exists v \neq 0$ $Av = \lambda v \quad \forall \lambda$ Eigenwert

v Eigenvektor

A def. posit. $\Rightarrow \sigma(A) \subset \mathbb{R}^+$

(a) gilt $-2 \notin \sigma(A)$

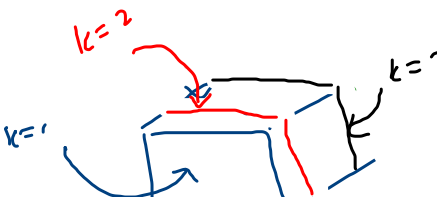
(b) es gibt jede matrix

symmetrisch ($A^T = A$) $\Leftrightarrow$ diagonalisierbar.

(c) $\sigma(A) \subset \mathbb{R}^+$ $\det(A) = \prod_{k=1}^n \lambda_k > 0$ (Ciarlo)
 $\lambda_k > 0 \quad \forall k$

(d) $A \Leftrightarrow$ regulär (i.e., invertierbar) $(\Leftrightarrow) \det(A) \neq 0$
(Ciarlo)

(10) $A \in \mathbb{R}^{5 \times 5 \times 5 \times 5}$ $A = (a_{ijkl})$ Responde (b)
Tensor dimensi 5
 $B \in \mathbb{R}^{n \times n}$ $B = (b_{ij})$ $\parallel$ $i+j+k+l$ rango 4
 (i, j, k, l)

$C \in \mathbb{R}^{3 \times 3 \times 3}$ $C = (c_{ijk})$

 c_{ij2}

$$C_1 = (c_{ij1})$$

$$C_2 = (c_{ij2})$$

$$C_3 = (c_{ij3})$$

C es de
 rango 3 i, j, k

dimensi 3

• $A \in \mathbb{R}^{\underline{n} \times \underline{n}}$ $A = (a_{ij})$

$$\begin{array}{ccc} a_{11} & \dots & a_{1n} \\ & \ddots & \\ & & a_{nn} \end{array}$$

Come tensor es de rango 2 (i, j)

dimension n

• $v \in \mathbb{R}^n$ rango 1 & dimension n

$v = (v_j)$. Un escalar es de rango 0 & dim 1

$$\textcircled{11} \quad f: \mathbb{R}^{\textcircled{2}} \rightarrow \mathbb{R}$$

$$f(x, y) = 1 - x^2 - y^2$$

pts crit

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = \left(\underline{-2x}, \underline{-2y} \right) \xrightarrow{\downarrow} (0, 0)$$

$$\Rightarrow x = y = 0 \quad \therefore (0, 0) \text{ is a pt crit}$$

$$H(f)(0, 0) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{(0, 0)} = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} \quad \sigma(H(f)(0, 0)) = \{-2\}$$

$\therefore H$ is def neg.

por tanto $(0,0)$ es un máximo relativo (C)

$$t = \pi(1 - x^2) \quad dt = -2\pi x dx$$

$$x = 0 \rightarrow t = \pi$$

$$x = 3 \rightarrow t = -8\pi$$

$$\textcircled{14} \quad 0 = \int_0^3 x \cos(\pi(1 - x^2)) dx \quad \downarrow \quad = -\frac{1}{2\pi} \int_0^3 \underbrace{-2\pi x}_{dt} \underbrace{\cos(\pi \underbrace{(1 - x^2)}_t)}_{\substack{0 \\ \uparrow}} dx$$

$$\downarrow \quad = -\frac{1}{2\pi} \int_{\pi}^{-8\pi} \cos t dt = \frac{1}{2\pi} \int_{-8\pi}^{\pi} \cos t dt = \frac{1}{2\pi} \left[\sin t \right]_{-8\pi}^{\pi} = \frac{1}{2\pi} \left(\cancel{\sin \pi} - \cancel{\sin(-8\pi)} \right) = 0$$

$$(15) \quad f: \mathbb{R} \rightarrow \mathbb{R}^2 \quad f(t) = (f_1(t), f_2(t))$$

$$g: \mathbb{R}^2 \rightarrow \mathbb{R} \quad \text{+7} \quad f(0) = \underline{(1, 0)}, \quad f'_1(0) = -2; \quad f'_2(0) = 1$$

$$\frac{\partial g}{\partial x}(\underline{(1, 0)}) = 1 \quad \& \quad \frac{\partial g}{\partial y}(\underline{(1, 0)}) = 4. \quad \text{Calculate } (g \circ f)'(0) =$$

$$\frac{\partial g}{\partial x}(\underline{f(0)}) \cdot f'_1(0) + \frac{\partial g}{\partial y}(\underline{f(0)}) f'_2(0) = 1 \cdot (-2) + 4 \cdot 1 = 2.$$